Task 1:

Five chunking strategies were evaluated against the WebMD Chronic Migraine PDF using three retrievers (Base Vector, AutoMerging, and BM25), producing 15 configurations in total. Two separate evaluation approaches were used, and it is important to understand what each can and cannot tell us. The first approach used a fixed set of 14 questions from eval_questions.txt, evaluated by an LLM judge on Faithfulness and Relevancy. Because the same questions were posed to every strategy, this is a genuinely fair cross-strategy comparison. The second approach computed MRR, Hit Rate, Precision, and Recall using QA pairs generated separately from each strategy's own nodes. This is where a critical methodological limitation arises: questions generated from Baseline (128-token) nodes are phrased to match 128-token chunk boundaries, and questions generated from Sentence Window nodes are phrased to match sentence-level context. Each strategy is effectively being tested on questions tailored to its own chunks, not a neutral benchmark, so MRR and Hit Rate figures are not directly comparable across strategies. One might propose fixing this by generating all QA pairs from a single canonical set of nodes, but this introduces the same bias in the opposite direction: questions generated from any one strategy's nodes would favor that strategy's retrieval. Furthermore, MRR requires a ground truth node ID for each question, and a node ID from the Sentence Window index has no direct equivalent in the Baseline index, so there would be no valid ground truth to evaluate against for the other strategies.

The LLM-judge fair comparison on the 14 fixed questions reveals a nuanced picture. For Base Vector, Sentence Window actually performed worse than Baseline, with Faithfulness dropping from 0.929 to 0.857 and Relevancy dropping from 0.786 to 0.714. For AutoMerging, Sentence Window improved Faithfulness from 0.857 to 0.929 and Relevancy from 0.643 to 0.786. For BM25, Sentence Window showed the largest regression: Faithfulness fell from 1.000 to 0.786 and Relevancy dropped from 0.571 to 0.500. Averaged across all three retrievers, Sentence Window’s average Faithfulness was 0.857 compared to 0.929 for Baseline, and average Relevancy was 0.667 for both strategies. The best strategy by average Relevancy in the fair comparison was Large (512/100) at 0.833, not Sentence Window. On the per-strategy retrieval metrics, Sentence Window achieved the highest within-strategy scores: MRR of 0.653 with BM25 and 0.578 with AutoMerging, compared to Baseline peaks of 0.535 and 0.288 respectively, and AutoMerging Hit Rate rose from 0.347 to 0.745. As noted above, these figures should be read as within-strategy indicators only.

Given the full picture, the selection of Sentence Window with BM25 rests primarily on its substantially stronger per-strategy retrieval metrics rather than on the fair comparison results, which actually favour Baseline on Faithfulness and Large (512/100) on Relevancy. The rationale is that retrieval accuracy, even when measured under the methodological constraints described above, reflects a more fundamental property of the pipeline: whether the correct content is being located at all. Even the best LLM cannot produce a correct answer if the wrong chunk is retrieved. The fair comparison, being limited to 14 questions with binary 0/1 LLM scoring, is subject to considerable judge variance on a small sample, whereas MRR and Hit Rate are measured over hundreds of QA pairs. Sentence Window's consistent advantage on these retrieval metrics across all three retrievers, despite the within-strategy caveat, provides sufficient justification for its selection. The 149 nodes produced by this configuration were exported as optimized_nodes.pkl for downstream use.

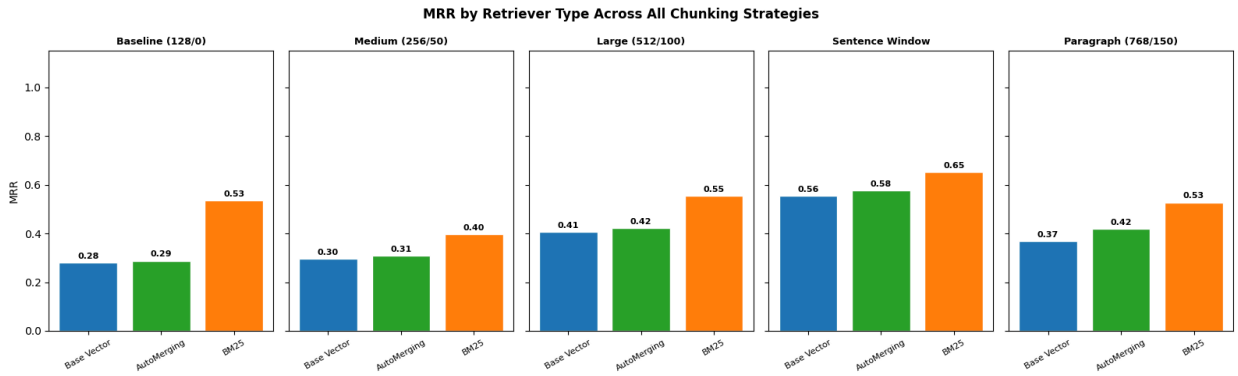

Fig 1.1 — MRR by retriever across all five chunking strategies (per-strategy QA pairs; not directly cross-comparable, see analysis).

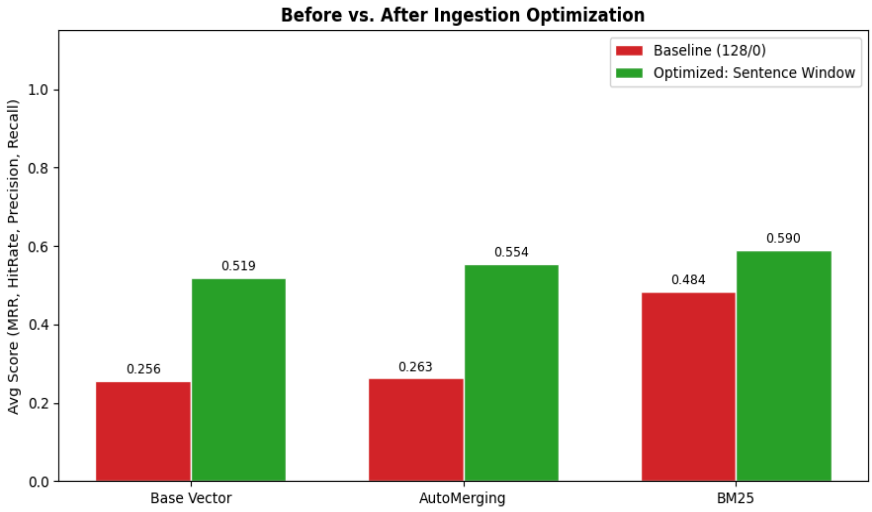

Fig 1.2 — Before vs. After average retrieval score (MRR, Hit Rate, Precision, Recall) across all retrievers (per-strategy QA pairs).

Task 2

A set of 13 questions was designed across five categories to stress-test the pipeline from multiple angles: Cross-Sectional (3 questions requiring synthesis across document sections), Keyword-Heavy (3 questions relying on exact medical terminology), Semantic/Contextual (3 questions requiring conceptual understanding), Structured Data (2 questions targeting list-formatted content), and Image-Based (2 questions whose answers exist only in figures). All questions were run through Base Vector, AutoMerging, and BM25 retrievers built on the Task 1 Sentence Window nodes. AutoMerging emerged as the strongest retriever for this diverse set, achieving the highest average Faithfulness (0.923, tied with Base Vector) and the highest Relevancy (0.769). This is a reversal from Task 1 where BM25 dominated on per-strategy retrieval metrics, explained by question diversity: cross-sectional and semantic questions require conceptual synthesis that AutoMerging handles better by merging related child nodes into larger context windows.

The category breakdown reveals a clear performance gradient. Structured Data scored highest (Relevancy 1.000) because the WebMD document explicitly enumerates medication categories in list form, making retrieval precise and answers directly verifiable. Semantic/Contextual also performed well (Relevancy 0.778), confirming that the Sentence Window strategy's broader context nodes support multi-sentence reasoning. Cross-Sectional and Keyword-Heavy both scored lowest on Relevancy (0.667), for different reasons: Cross-Sectional questions require synthesizing information scattered across

sections that a single retrieved node cannot fully capture, while Keyword-Heavy questions showed uniform performance across all three retrievers, suggesting that the sentence window context was sufficient for keyword matching but not enough for full answer coverage.

The Image-Based category produced the lowest scores of any category, with average Faithfulness and Relevancy both at 0.500 across all three retrievers. This is expected: the text-only pipeline cannot access content stored in the PDF's vector-drawn infographics, so answers to image-dependent questions were either hallucinated from general knowledge or correctly declined. The uniform 0.500 score across all retrievers confirms that this is a fundamental limitation of the pipeline rather than a retriever-specific failure. This highlights a clear gap in text-only RAG pipelines for PDF documents with embedded visuals, which would require multimodal ingestion or structured image captioning to address.

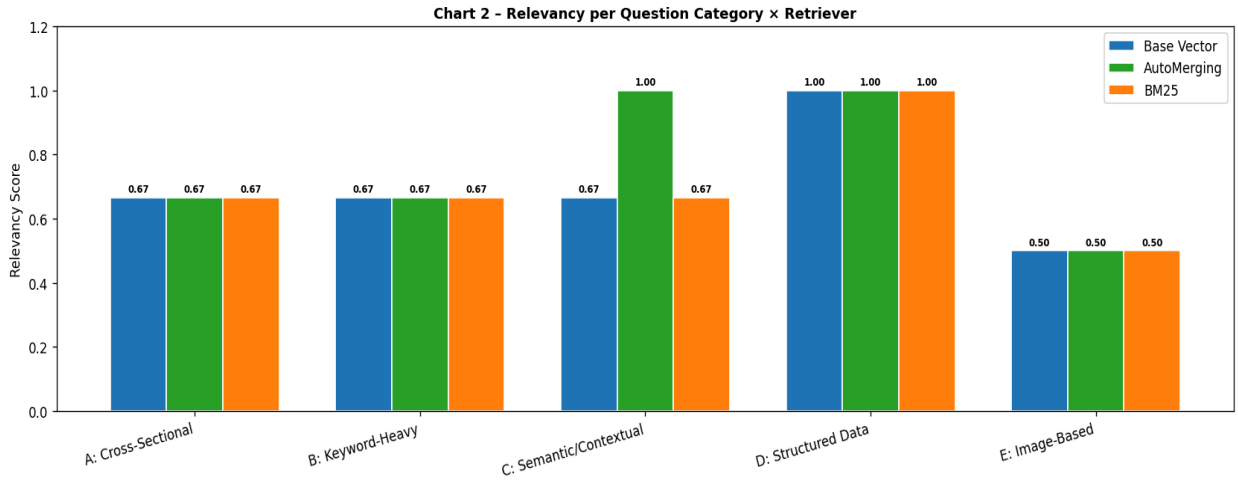

Fig 2 — Relevancy per question category by retriever. Structured Data (D) scores highest; Image-Based (E) reveals the text-only RAG blind spot.

Task 3:

Pages 5 and 6 of the WebMD PDF contain vector-drawn infographics (the migraine stages diagram and the comorbidities chart) that are completely invisible to standard text extraction. A hybrid PyMuPDF strategy was implemented to address this. For most pages, embedded raster images were extracted directly using get_images(), yielding four images across pages 1, 2, 3, and 9 that contained only photos and logos with no readable text. Pages 5 and 6 presented a different challenge because their visual content is entirely vector-drawn: get_images() returned only thousands of tiny decorative elements. For these pages, the entire page was rasterized at 200 DPI using get_pixmap() and saved as a full-page PNG. Tesseract OCR was then applied to all eight extracted images, and only the two full-page renders yielded readable content: 1,316 characters from the page 5 stages diagram and 1,898 characters from the page 6 comorbidities infographic, totalling 3,327 characters once assembled into a single OCR document with labelled sections.

The extracted text was chunked using the same Sentence Window parser as the Task 1 optimized strategy, producing 16 OCR nodes that were merged with the existing 149 nodes to form a combined index of 165 nodes. Three retrievers were rebuilt on the combined index, with BM25 top_k increased from 3 to 5 to allow OCR nodes to compete against high-frequency text nodes in keyword ranking. The most compelling evidence of OCR impact comes from the Image-Based questions in Task 2. Before OCR integration, BM25 retrieved only text nodes and the model hallucinated phase descriptions from general knowledge. After OCR integration, BM25 correctly retrieved the page 5 OCR node with a score of 2.298 and produced a fully accurate answer: prodrome lasting a few hours to a few days, aura lasting 5 to 60 minutes, headache or attack lasting 4 to 72 hours, and postdrome lasting 24 to 48 hours.

OCR integration produced mixed results across the three retrievers. Base Vector showed the clearest improvement, with Relevancy rising by +0.231 (from 0.538 to 0.769) while Faithfulness held steady. BM25 gained +0.077 on Faithfulness (from 0.769 to 0.846) with Relevancy unchanged. However, AutoMerging experienced a Relevancy regression from 0.769 to 0.538 (−0.231), likely due to single-run LLM judge variance rather than a systematic degradation, since Faithfulness remained at a perfect 1.000. BM25 showed the largest qualitative improvement by correctly answering the migraine stages question that it had previously hallucinated. The combined 165-node index was exported as optimized_nodes_with_ocr.pkl for use in Tasks 4 and 5.

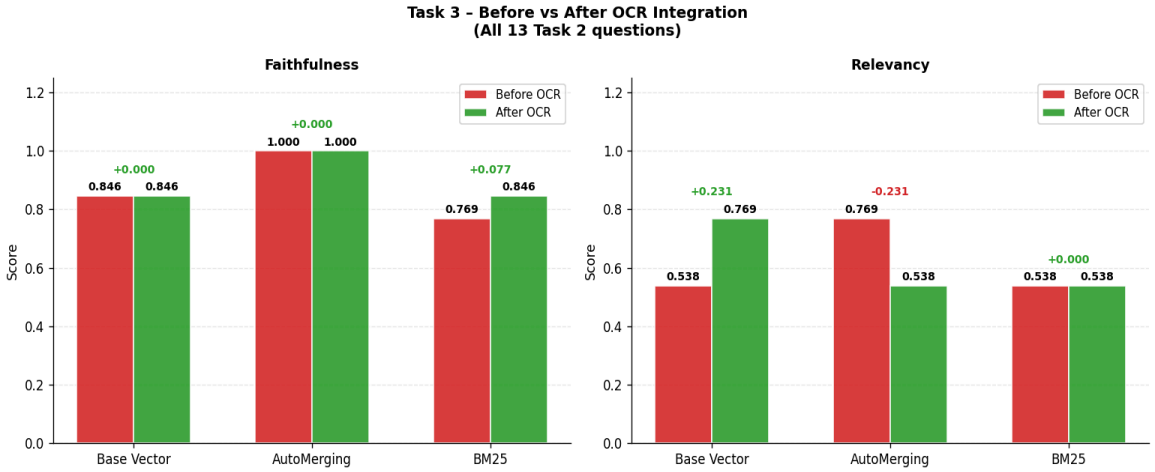

Fig 3 — Before vs. After OCR integration: Faithfulness and Relevancy across all retrievers (13 Task 2 questions).

Retriever	Before (Faith. / Relev.)	After (Faith. / Relev.)	Delta
Base Vector	0.846 / 0.538	0.846 / 0.769	+0.000 / +0.231
AutoMerging	1.000 / 0.769	1.000 / 0.538	+0.000 / -0.231
BM25	0.769 / 0.538	0.846 / 0.538	+0.077 / +0.000

Table 3 — Faithfulness and Relevancy before vs. after OCR integration (single-run results).

Task 4:

A weighted hybrid retriever was implemented using LlamaIndex's QueryFusionRetriever in dist_based_score mode, combining a 0.6-weighted vector retriever and a 0.4-weighted BM25 retriever. Both component retrievers ran at top_k=10 internally to give the fusion step a wide candidate pool, with the final output trimmed to top_k=5. Retrieval-level metrics (MRR, Hit Rate, Precision, Recall) were evaluated on a 149-question QA dataset auto-generated from the clean text nodes, while generation-level metrics (Faithfulness, Relevancy) used the same 13 hand-crafted Task 2 questions with an LLM judge. These two evaluation tracks are separated by necessity. Retrieval metrics like MRR require a ground-truth node ID for each question: the evaluator must know which specific chunk is the correct answer so it can measure where that chunk appears in the ranked results. The 13 Task 2 questions were crafted to test end-to-end answer quality across diverse categories, not to point at a specific node, so they lack the node-level labels MRR requires. Conversely, the auto-generated QA pairs are created directly from individual nodes, giving each question a known correct node ID, but they are narrowly scoped to the content of a single chunk and cannot test cross-sectional reasoning or answer quality the way the hand-crafted questions can. The two approaches are therefore complementary: auto-generated QA pairs measure retrieval precision at scale, while the Task 2 questions measure whether the full pipeline produces useful answers to realistic queries. The hybrid retriever achieved the highest MRR (0.692, compared to 0.651 for BM25 and 0.553 for Base Vector and AutoMerging) and the highest Hit Rate

(0.805, compared to 0.745 for BM25). These gains confirm that fusing semantic and keyword signals surfaces relevant chunks that neither method finds alone. Precision appears low across all retrievers (0.146–0.161), but this is a mathematical artifact of the evaluation setup rather than a quality problem: each QA pair has exactly one ground-truth node, so with top_k=5 a perfect retriever would score at most 1/5 = 0.200. Precision is simply Hit Rate divided by top_k (e.g., Hybrid: 0.805 / 5 = 0.161), making MRR and Hit Rate the more informative metrics in this sparse-label setting.

On generation quality, the Hybrid retriever matched the single-retriever baselines rather than surpassing them, with Faithfulness of 0.923 and Relevancy of 0.692. AutoMerging remained the leader on Faithfulness (1.000) for this question set. This pattern indicates that retrieval accuracy was still improving with hybridization but that generation quality was approaching a ceiling constrained by the top-k context window passed to the LLM, rather than by which chunks were retrieved. The hybrid approach nonetheless represents the best retrieval foundation available, and it was carried forward as the baseline for Task 5.

The choice of a 0.6/0.4 weighting in favor of vector search reflects the nature of the document and question set. The WebMD PDF contains dense medical prose where semantic understanding is often more useful than exact keyword matching, particularly for cross-sectional and contextual questions. BM25 still contributes meaningfully for queries containing precise terminology such as VYEPTI or CGRP, where exact token matching is highly reliable. The weighted fusion balances both retrieval modes and consistently outperforms either method in isolation on MRR and Hit Rate.

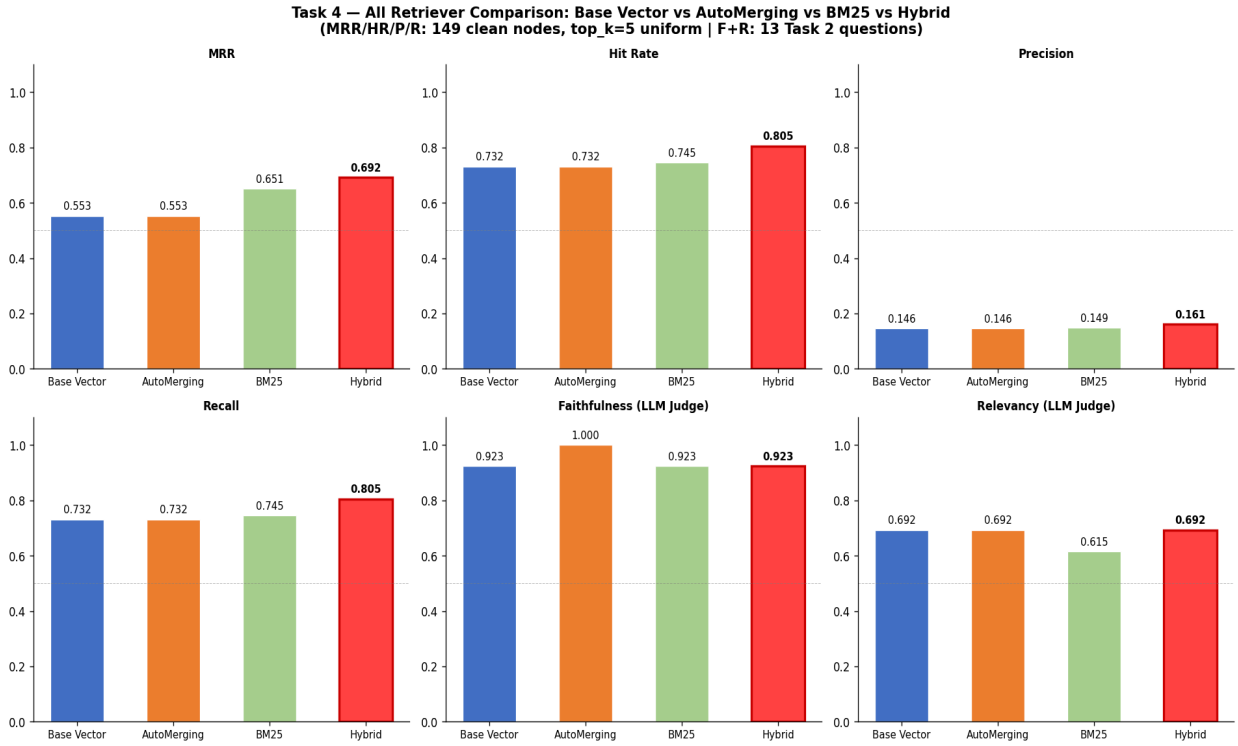

Fig 4 — All-retriever comparison across six metrics. Hybrid leads on MRR (0.692) and Hit Rate (0.805).

Task 5:

Three improvements were implemented on top of the Task 4 Hybrid baseline. The first was Multi-Query Expansion, configured through LlamaIndex's QueryFusionRetriever with num_queries=3, which prompts the LLM to generate two additional phrasings of each query. All three variants are run through retrieval and fused with Reciprocal Rank Fusion (RRF). The motivation was to close the retrieval recall gap (Task 4 MRR of 0.692 means the correct chunk is not ranked first roughly 30% of the time). In

practice, on this single-domain medical corpus, LLM paraphrases reused the same vocabulary as the original query, so Hit Rate and Recall were unchanged and MRR fell slightly to 0.660 due to RRF score normalization. Its main contribution was widening the candidate pool to top_k=10 for the downstream reranker. The second improvement was Cross-Encoder Re-ranking: a SentenceTransformerRerank postprocessor using cross-encoder/ms-marco-MiniLM-L-6-v2, which scores each of the 10 candidates by reading the query and passage jointly before selecting the top 5 passed to the LLM. This is architecturally superior to cosine similarity, which scores query and passage independently and cannot model their interaction.

The third improvement was Adaptive Semantic Deduplication, a novel contribution implemented as a custom SemanticDedupPostprocessor that plugs into the LlamaIndex BaseNodePostprocessor interface. It uses greedy cosine deduplication to remove candidate nodes whose similarity to any already-accepted node exceeds a tunable threshold, ensuring that the final context window passed to the LLM is diverse rather than redundant. A sensitivity sweep at thresholds from 0.85 to 0.95 showed that the RRF fusion step pre-diversifies the candidate pool so thoroughly that deduplication removed at most one node per query at every threshold tested. This validates that deduplication is most impactful on single-query retrievers rather than multi-query fusion pipelines, and it was included as a principled architectural component even though its marginal contribution in this specific setup was small.

Evaluation used 3-run averaging with an LLM judge, reducing per-run variance from approximately plus or minus 7.7% to plus or minus 4.4%. The Multi-Query plus Reranker pipeline achieved Faithfulness 1.000 and Relevancy 0.923 with zero run-to-run variance across all three evaluation runs, compared to the Task 4 baseline of Faithfulness 0.897 (plus or minus 0.044) and Relevancy 0.846 (plus or minus 0.077). Adding semantic deduplication on top of this pipeline did not improve scores further (Faithfulness 0.974, Relevancy 0.846), because the RRF-fused pool was already diverse enough that deduplication rarely activated. The recommended final pipeline is therefore Multi-Query Expansion combined with Cross-Encoder Re-ranking, representing gains of +0.103 Faithfulness and +0.077 Relevancy over the Task 4 Hybrid baseline.

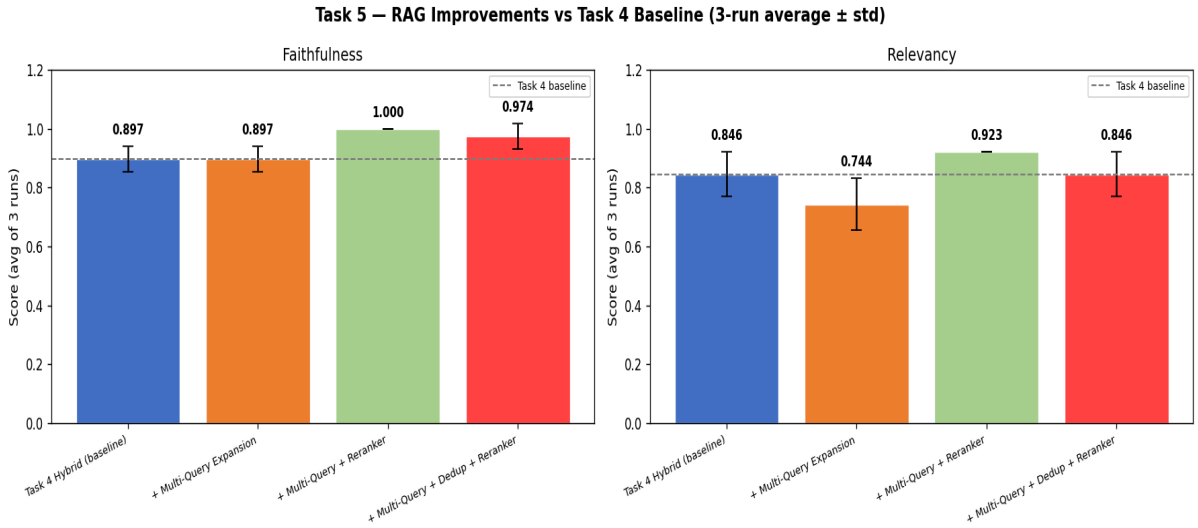

Fig 5 — Task 5 pipeline comparison (3-run avg, with standard deviation). Multi-Query + Reranker achieves Faithfulness 1.000 and Relevancy 0.923 with zero variance.

Pipeline	Baseline (Faith. / Relev.)	Result (Faith. / Relev.)	Delta
Task 4 Hybrid (baseline)	0.897 / 0.846	0.897 / 0.846	baseline
+ Multi-Query Expansion	0.897 / 0.846	0.897 / 0.744	+0.000 / -0.103

+ Multi-Query + Reranker	0.897 / 0.846	1.000 / 0.923	+0.103 / +0.077
+ Multi-Query + Dedup + Reranker	0.897 / 0.846	0.974 / 0.846	+0.077 / +0.000

Table 5 — Task 5 pipeline results. Multi-Query + Reranker is the recommended final configuration.